

User Control in Tourism Recommender Systems

Group W3

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Chair of Connected Mobility

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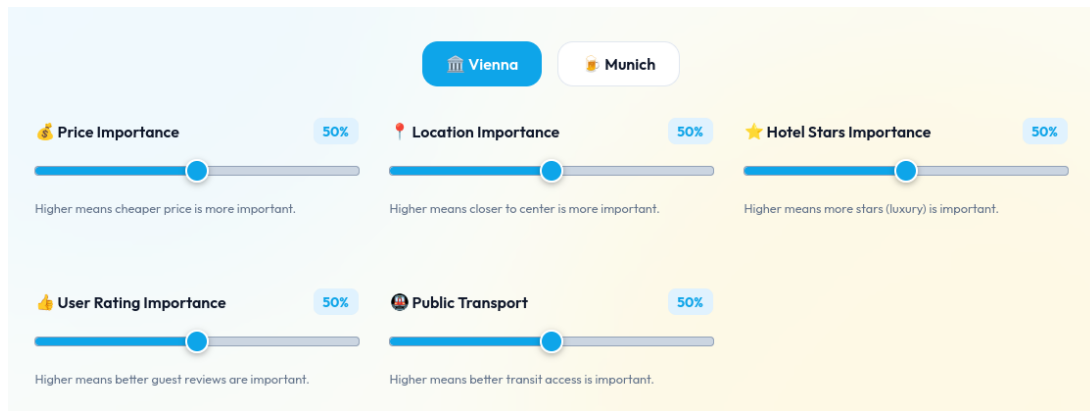
TUM Uhrenturm

Project Overview: Group W3

- **Topic:** User Control and Strategies for Influencing Algorithms.
- **Core Goal:** Moving away from “black-box” systems by giving users direct influence over ranking logic [4].
- **Motivation:**
 - Improving user trust through transparency [1].
 - Compliance with the **EU Digital Services Act (DSA)** (Requirement for transparent and explainable algorithms).
 - Allowing users to express complex, multi-dimensional preferences [3].

Current Status: Static Prototype Planning

- **Frontend Plan:** React¹ + Vite² + CSS.
- **UI Components Planned:**
 - **City Selector:** Support Vienna and Munich.
 - **Preference Sliders:** Five interactive sliders (Price, Location, Stars, Rating, Transport) [1].
 - **Results Grid:** Real-time visualization with Match Percentage badges.



¹<https://react.dev/>

²<https://vite.dev/>
Group W3 Gallob (TUM)

Proposed Algorithms for Comparison

Aim: allow users to choose between two distinct algorithmic approaches [2]:

Algorithm A: Weighted MAUT

- Linear combination of normalized scores [4].
- 100% transparent; users see exact contributions.

Algorithm B: Constraint-Based / Critiquing Hybrid

- Hard limits (e.g., “Max 150€”) + “More like this” critiques [1].
- High transparency through explanation of satisfied constraints.

Algorithms Deep Dive

Algorithm A: Multi-Attribute Utility Theory (MAUT)

- quantifies personal utility for an outcome [4].
- Importance weights specify tradeoffs between competing attributes [4].
- Overall utility $U(X)$ is calculated using an additive value function: $U(X) = \sum_{i=1}^5 w_i \cdot u_i(x_i)$ [4].

Algorithm B: Constraint-Based & Critiquing

- Express desired properties of an “ideal result” via constraints [4].
- Unlike MAUT, can exclude items that do not fit conditions (e.g., price limit) [4].
- returns logically sound results, because of user input (no guessing required) [1].
- Easy explicable and transparent for users [4, 1].

Key Features & Next Steps

- **Key Feature:** Selection menu to switch between “Manual Weighting” and “Constraint-Guided” modes.
 - **Transparency Panel:** Detailed view of mathematical calculations to improve transparency [3].
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1. **Algorithm Implementation:** Implement the algorithms and the respective choice fields.
 2. **Algorithm Selector UI:** Add the toggle to the main dashboard.
 3. **Explanation:** Implement detailed explanation of the calculations as per DSA requirements.

Thank you for your attention!

Questions?

References I

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- [4] Christian Schmitt, Dietmar Dengler, Mathias Bauer u. a. „The maut-machine: an adaptive recommender system“. In: *Proceedings of the ABIS Workshop, Hannover, Germany*. 2002.